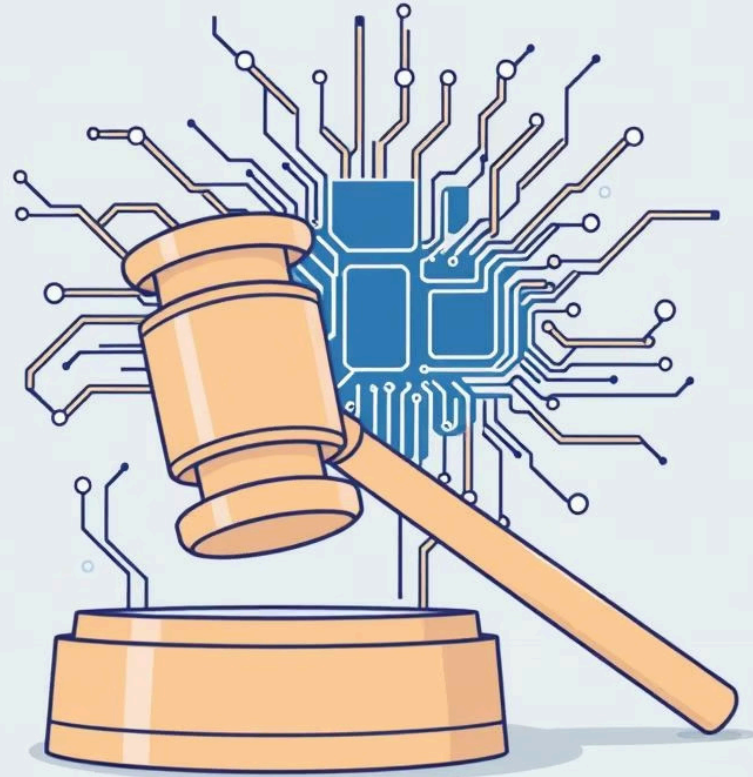




ADAL - AI Driven Analysis for Law

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Department: CS

Tagline: “AI-powered precedent intelligence for Pakistan’s justice system”

Presentation Outline

- 1 The Problem: Crushing the Justice System
- 2 The Solution: ADAL
- 3 Competitor Analysis
- 4 Scope of the Project
- 5 Core Features
- 6 Impact of ADAL
- 7 Timeline
- 8 Tools and Techniques
- 9 Work Division



The Problem: Crushing the Justice System

Pakistan's legal system faces significant challenges, from overwhelming case backlogs to inefficient research methods, all contributing to a delay in justice.



Case Backlogs & Overburdened Courts

- Over 2.2 million pending cases nationwide.
- Supreme Court judgments can take months to finalize.
- Judges handle hundreds of cases daily, exceeding human capacity.



Inefficient Legal Research

- Lawyers and judges spend hours searching bulky PDFs, digests, and physical libraries.
- Finding the right precedent is like finding a needle in a haystack.



Limited Access to Precedents

- No centralized, AI-driven repository for Pakistani case law.
- Current keyword-based tools miss crucial context and semantic similarity.



Delayed Justice Delivery

- Slow legal research directly causes delayed judgments and inconsistent decisions.
- The unfortunate outcome: **Justice delayed is justice denied.**

The Solution: ADAL

An AI-powered research and decision-support system tailored for Pakistan's legal ecosystem.



Efficiency Boost

Addresses case backlogs by cutting research time from hours to minutes.



Insightful Precedent Analysis

Provides precedent-driven insights to strengthen legal arguments.



Accessible Justice

Ensures faster, more consistent, and accessible justice for lawyers, judges, and the courts.

Bottom Line:

Not just another chatbot — a professional-grade AI assistant that transforms how legal research and judicial decision-making are done in Pakistan.

Competitor Analysis

ADAL offers a comprehensive, professional-grade legal AI solution, specifically designed for judges and lawyers, addressing critical gaps in existing Pakistani offerings.

Feature / Tool	Bahaq Legal AI	LawScript	PakLawAssist / LawGPT	Our Platform
Precedent Finder (RAG)	No	Limited (semantic search only)	No	Yes (semantic + similarity ranking)
Case Similarity Search	No	No	No	Yes
Argument Suggestions	No	No	No	Yes (precedent-backed, for/against)
Multi-Document Comparison	No	No	No	Yes
Decision Support Insights	No	No	No	Yes (probability & support tool)
Target Users	Citizens	Lawyers	Citizens & Lawyers	Judges + Lawyers (professional-grade)

ADAL distinguishes itself with advanced capabilities like precedent ranking, argument suggestions, and decision support tools, making it an indispensable legal intelligence platform for professionals.

Scope of the Project

- 1 Build an **AI-powered legal assistant** specialized for Pakistani courts.
- 2 Support for both **lawyers & judges**.
- 3 Provide localized precedent analysis, summarization, and decision support.
- 4 Designed as a **scalable platform** (modular system with clear phases).

4.

Features Bundled – Core Functionalities

ADAL's comprehensive features streamline legal workflows, from document analysis to research and drafting.



Document Processing & Accuracy

- OCR Integration for scanned documents.
- Accurate Legal Parsing to Ensure correct extraction of sections, clauses, and articles
- Citation Checker validates references.
- Authenticity Verification for legal texts.



Legal Knowledge & Research

- Auto-Insert Precedents with relevance.
- Advanced Search Filters for precision.
- Auto-Update Case Law to stay current.



Drafting & Export Tools

- Smart Draft Assistant for rapid document creation.
- Multi-Format Export (Word, PDF, etc.)

Features Bundled – Core Functionalities

ADAL's comprehensive features streamline legal workflows, from document analysis to research and drafting.



Collaboration & Workflow

- Editing & Collaboration Mode
- Case Timeline Visualizer



Multilingual Support

- English
- Urdu



Advanced Legal Intelligence

- Keyword & Theme Extraction
- Similarity Heatmap
- Contradiction Detection Across Documents
- Judge Analytics
- Legal Chat Assistant

Impact of ADAL: Transforming Justice

ADAL is designed to create a ripple effect, bringing significant benefits to every stakeholder within Pakistan's legal ecosystem, from individual citizens to the system as a whole.



For Judges

- Faster access to relevant precedents
- Reduced case backlog pressure
- More consistent decision-making



For Lawyers

- Saves hours of manual research
- Stronger, evidence-backed arguments
- Enhanced case preparation



For Citizens

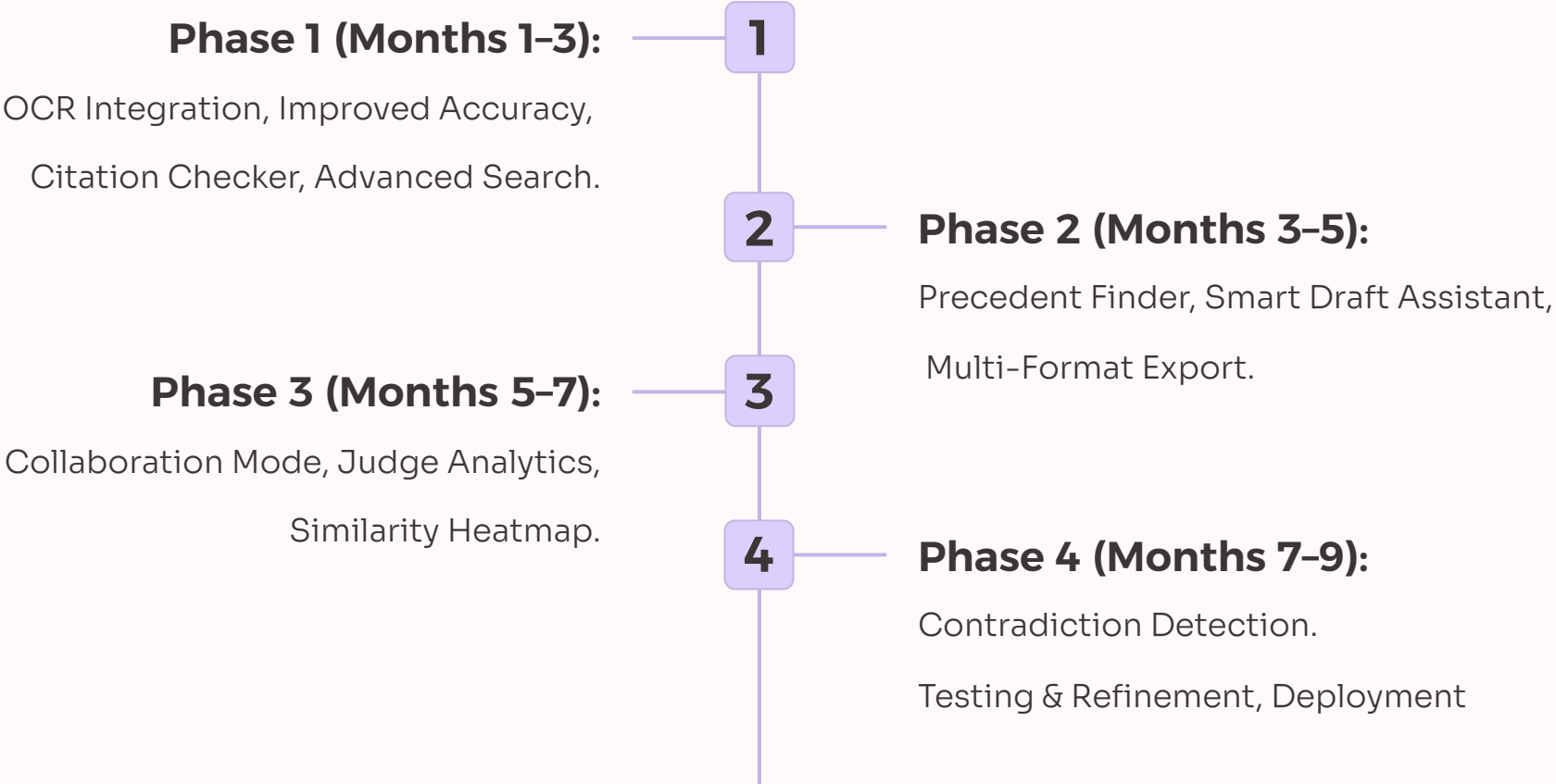
- Faster case resolution
- Improved access to justice
- Increased transparency in the legal process



For the Legal System

- Scalable digital infrastructure
- Step toward modernization & AI adoption
- Foundation for future innovations

Timeline



Tools & Techniques

AI & NLP Models:

- Transformers (BERT/LegalBERT)
- LangChain
- RAG pipelines
- Embeddings & Similarity: OpenAI / HuggingFace

Backend:

- Python (FastAPI / Flask for APIs)

Frontend:

- React.js (modern UI for lawyers/students)

Database:

- PostgreSQL / MongoDB (for storing documents & metadata)

OCR:

- Tesseract / EasyOCR (scanned docs → text)

Deployment:

- Docker (Local)

Work Division



Hasan Ali

Data & Preprocessing

- Collecting judgments, laws, and legal documents
- Cleaning, structuring, and annotating datasets
- Building knowledge base for RAG



Hasnain Ali Arshad

AI Model Development

- Case summarization (LLM fine-tuning)
- Precedent finder & similarity search
- Law/section extraction & NLP pipelines



Abdullah Azeem

Evaluation & Deployment

- Testing accuracy & reliability of AI outputs
- Integrating AI modules with frontend/backend
- System deployment, scalability, and optimization

ADAL

Revolutionizing Justice Through AI

Thank you. Any questions?

Appendix: Key Datasets

Our AI models are trained and refined using comprehensive legal datasets, ensuring accuracy and relevance to the Pakistani legal context. Below are examples of the types of datasets foundational to ADAL's capabilities.

Dataset Name	Description / Link
Pakistan Supreme Court Judgments Archive	Comprehensive collection of decisions from the Supreme Court of Pakistan. www.supremecourt.gov.pk/judgement-search/
Pakistan Statutes and Ordinances	Digitized repository of all federal and provincial laws, acts, and ordinances. https://pakistancode.gov.pk/english/LGuOxBD.php
Legal Literature and Commentary Corpus	A curated collection of legal articles, books, and expert commentaries relevant to Pakistani law. https://journals.au.edu.pk/ojsrcr/index.php/crc/article/download/259/155/369